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An Examination of Context-Aware Augmentative and Alternative Communication with Automated Response Options for Academic Communication from Autistic Children

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Abstract

Children with intellectual and developmental disabilities (IDD) who use augmentative and alternative communication (AAC) often face barriers to full academic participation due to the difficulties of navigating traditional AAC displays. Advances in artificial intelligence may enable context-aware AAC tools that generate response options based on linguistic input from communication partners, enabling faster and more effective communication and potentially increasing academic participation. This study compared the effects of a context-aware AAC prototype to traditional AAC in supporting academic communication for three autistic preschool children (two male; one female) using an ABAB reversal design. Findings suggest that the context-aware AAC resulted in more frequent, accurate, and faster communication responses from all participants. Although this study used a prototype, results indicate that context-aware AAC technologies may enhance academic communication and participation for young autistic children. Further research is needed to explore the broader impact and long-term effectiveness of such tools in academic settings.

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This research was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and received approval from the University of Arkansas Institutional Review Board (protocol approval #1709063747).

Ethical Considerations

Informed written consent for participation and publication was collected from the parents/guardians of all participants prior to beginning this study. This study was approved by the second author's Institutional Review Board (protocol approval #1709063747).

Declaration of Conflicting Interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Keywords

AAC; context-aware; autism; academic

Autistic children or children with other intellectual and developmental disabilities (IDD) who have complex communication needs are faced with many challenges that may restrict their abilities to develop language skills and actively participate in community activities, health care, family life, and educational settings (Holyfield, Light et al., 2025; Light et al., 2025). The goal of augmentative and alternative communication (AAC) has consistently been to support communication, language development, and engagement in meaningful interactions for children with complex communication needs (Holyfield, Light et al., 2025; Light et al., 2025). AAC systems can include a wide range of supports from picture cards or communication boards to high-tech options such as speech-generating devices or mobile devices with communication apps (Ganz et al., 2023). Although the specifications of these systems continue to evolve as technology develops, the goal of empowering people to express themselves and be heard remains a constant.

In light of this goal, it is imperative to consider the limitations of current systems and design technologies that may be able to provide more effective and efficient support for individuals who require AAC. In a recent study, Lorah and colleagues (2024) interviewed adult AAC users with developmental disabilities about their experiences and found that one of the prevailing challenges was the cumbersome nature of existing technologies and the associated slow rate of communication. Participants reported that during fast-paced conversations, communication partners would sometimes become inattentive in the time it took the AAC user to input their response into the device or the conversation would move on before their response was formed (Lorah et al., 2024).

Due to the difficulty of interviewing preschoolers who use AAC, researchers often rely on teacher and caregiver reports to identify barriers. Similar to reports from adult AAC users, caregivers and educators describe significant difficulties. Parents often report that the features and functions of AAC do not align with their child's individual needs. One major challenge is that locating and selecting desired vocabulary for effective communication can be overly time-consuming (Berenguer et al., 2022). In particular, parents of preschool AAC users struggle with organizing the symbols on the device, choosing an appropriate graphic system, and learning navigational principles. They also express having inadequate time to support their child's AAC use during everyday activities (Stadskleiv, 2017). Teachers likewise identify challenges with the way AAC systems are organized (Leatherman & Wegner, 2022). Traditionally, high-tech AAC systems are organized with a series of topic pages (folders) of symbol icons in a grid layout. While organization is needed to house vocabulary on an AAC device, this organization often yields several navigational steps and selections for one word. As a result, teachers must spend time teaching AAC users to navigate through a complex maze of folders and pages, which cuts into academic instruction time (Leatherman & Wegner, 2022). Further, this results in children with developmental disabilities who use AAC expending significant time and effort participating in interactions

as a best-case scenario and being left behind as interactions move forward without them as a worst-case scenario (Quinn et al., 2023).

This challenge in icon organization highlights the importance of context in communication interactions. Especially during early language learning, communication interactions tend to focus on the immediate environment or context, such as features or motions of the communicator, features of their partner, what is happening around them, the responses of the partner, and the location or setting (Holyfield et al., 2019; 2024). When there is a disconnect between the communication context and the layout of the AAC, it places a greater burden on both the AAC user and the communication partner. The AAC user must filter out non-contextual information and sort through potentially irrelevant vocabulary, which increases the cognitive load required for communication and can slow down the communication interaction (Cai et al., 2023). Meanwhile, the communication partner must stay engaged with the AAC user and allow time for a response. This may be especially challenging during student/teacher interactions in a busy classroom environment, leading teachers to require high and continuous levels of support while working with students who rely on AAC to communicate (Da Fonte et al., 2022).

Previous research has examined ways to overcome the disconnect between the immediate context and the design of AAC through just-in-time (JIT) programming. JIT allows new vocabulary or phrases to be added to the AAC on the fly or “just-in-time” as opportunities arise during daily interactions (Holyfield et al., 2019). The goal of JIT is to ensure that communication partners can capitalize on a child’s interests and real-time activity by adding relevant vocabulary in the moment it is needed. However, when using traditional AAC, the task of adding new vocabulary generally falls to communication partners, who may need special training from a professional, such as a speech-language pathologist, in order to be successful (Light et al., 2025). Furthermore, teachers and other school professionals are typically engaged in other responsibilities which can make the multitasking of programming while leading an interaction difficult. Consequently, even solutions such as JIT programming that can streamline access to relevant vocabulary still require intentional work on the part of communication partners and for this reason may be infeasible or underutilized at times.

Therefore, researchers have more recently focused on ways to leverage technological advances to create new or improved AAC systems that may overcome these limitations (Holyfield, Light et al., 2025; Williams & Holyfield, 2025). In fact, AAC users themselves state a desire for technology to be able to better predict things they would like to communicate based on time of day, location, and the speech of people around them (Lorah et al., 2024). Artificial intelligence (AI) technology is advancing in ways that may begin to make this dream a reality (Cai et al., 2023; Holyfield, Light et al., 2025). For example, a “context-aware” AAC would be able to identify and respond to the user’s environment, such as interactions, communication partner’s speech, and location (Kane et al., 2012). This type of technology could encode communication partner speech into visual input on the device as the interaction is taking place, thus providing immediate and high-relevance communication support JIT without requiring programming from a communication partner (Holyfield et al., 2024; Holyfield, O’Neill Zimmerman et al., 2025).

Moreover, by empowering AAC users to anticipate and initiate communication that aligns with their goals, preferences, and daily contexts, such systems could also create meaningful opportunities to build self-determination (Raley et al., 2020). Self-determination includes abilities, skills, and attitudes that enable people to act as causal agents in their own lives, including making choices, setting goals, and advocating for themselves and others (Shogren & Raley, 2022). In particular, context-aware AAC has the potential to reduce the cognitive demands associated with locating vocabulary so that AAC users can devote their efforts to volitional acts, such as selecting among options, initiating communication, and persisting in goal-directed action. When the technology surfaces contextually relevant options in real time, it may increase opportunities for children to (a) experience successful, efficient communication (supporting perceived competence), (b) feel empowered about how and when they respond (supporting autonomy), and (c) participate more fully in school, home, and community interactions (supporting relatedness; Holyfield, O'Neill Zimmerman et al., 2025). For instance, users would be able to build greater autonomy in decision making, communicate choices more efficiently, and self-advocate in real time, all of which are core components of self-determined action.

While automation of JIT programming is promising, communication partners- such as teachers and parents- must remain critical consumers of AI. They should verify AAC outputs for accuracy and ensure the user's intent is not misinterpreted, while also understanding the limitations of these devices (Cai et al., 2022). Because this technology is still in development, research on its use with autistic children or children with other complex communication needs is only emerging. To date, two small studies have evaluated AAC with automated JIT programming with autistic children or with children with other developmental disabilities (Holyfield et al., 2024; Holyfield, O'Neill Zimmerman et al., 2025). Both studies found the approach to be favorable. However, both studies focused on participation in social routine interactions and therefore provide limited insight into the approach's application to supporting children in demonstrating academic knowledge.

Academic communication is essential to young children's participation in school as well as their ability to demonstrate their knowledge to teachers, peers, and others in the school setting. Many activities in school revolve around students' ability to answer the teachers' questions, and many teachers use this method to assess the students' academic skills and knowledge. Yet only a handful of examples of AAC intervention research focused on the communication of academic vocabulary exist (e.g., Griffen et al., 2025 Yorke et al., 2018). Further, research has yet to be completed to evaluate the effect of automated JIT programming on responses to academic questions.

Given this gap in the literature, the aim of this study was to expand the research on AAC intervention by targeting communication of academic knowledge in response to knowledge questions and to provide further evaluation of context-aware AAC with autistic children. The purpose of this study was to compare the effects of using a context-aware AAC prototype with automated response options to traditional AAC with a topic grid display on the academic communication of three young autistic children. The prototype offered real time choices for responses to participants by displaying the list of choices stated by the

communication partner. The academic knowledge the children communicated in the study was color knowledge, a prominent academic skill for young children.

Method

This study was approved by the Institutional Review Board of the second author's university before any participants were recruited (protocol approval number 1709063747). A guardian provided informed consent for their child's participation in the study.

Participants

Eligibility criteria for this study included (a) parent/guardian interest in child participation, (b) parent/guardian and school report of a documented diagnosis of autism, (c) participant age of four and five years, and (d) participant classification of emerging symbolic communicator (or someone who is beginning to use signs, gestures, pictures, or words to represent people, places, objects, and ideas). The age criteria of four or five years old was chosen because this represents a crucial period of language development and because these children needed increased opportunities to practice responding to academic questions as they prepared to enter formal schooling soon. Each participant's status as an emerging symbolic communicator was verified through professional report and screening prior to study enrollment. Three children, Anna, Blake, and Calvin (pseudonyms), participated in the study. Table 1 contains further demographic and assessment information about each participant.

They were recruited through teachers in the local community and an on-campus university program for autistic children. Guardians signed consent forms for all children participating in the study. All participants primarily utilized prelinguistic communication, and professionals reported that they all demonstrated limited participation in school activities, with little to no spontaneous communication within classroom activities or in response to questions from teachers. All children had at least some prior experience with high-tech AAC devices prior to the start of the study, but their experience levels varied. Anna had the most experience using high-tech AAC. She had consistent access to her own high-tech AAC device (a tablet with an AAC app), but her spontaneous use of it was primarily for requesting preferred activities (e.g., listening to a favorite song) and items (e.g., a favorite stuffed animal). Blake and Calvin in contrast had more limited prior experience with high-tech AAC. Neither of them had their own AAC device. However, both of them interacted using high-tech AAC during small portions of their school day, such as while with their school speech-language pathologists (SLPs). Blake and Calvin also had experience with low-tech AAC that was integrated throughout their classroom as a support for all students.

Setting and Context

The sessions took place during art-related activities that naturally incorporated a wide range of colors, making them an ideal context for this study. Colors are prominent in early childhood classrooms, and color identification is a key academic skill as indicated by its inclusion in the school readiness standards in the state where the study occurred. Because

participants were preparing to enter formal schooling, the ability to answer color-related questions was relevant for supporting engagement in academic activities. Art activities varied and were selected based on participants' lead. Example art activities included painting, coloring, and sticker books. Two settings were utilized based on participants' naturally occurring learning environments. All sessions with Anna were conducted in a quiet corner of a university autism program's playroom, with a small table for coloring and other activities. For Blake and Calvin, all sessions were conducted in a small playroom of a community school, with a similar activity table available to allow coloring interactions to occur.

Design

An ABAB reversal single-case experimental design (Ledford & Gast, 2024) was used to evaluate the impact of a context-aware AAC prototype in comparison with a popular traditional AAC option. The independent variable of the study was the AAC technology condition. Condition A, the traditional option, was the LAMP app with default programming open to the colors page (see Materials section below for more details). Condition B, the context-aware option, was a prototype in which four response options to a question appeared in near real-time as they were listed out by the communication partner – e.g., if the investigator asked, “Is the puppy blue, brown, green, or black?” blocks of each color appeared on the AAC device screen as each color was spoken aloud by the investigator, available for the participant to select from. All participants completed two phases in each condition, alternating across phases. Phase shifts occurred after a preset 5 trials in each condition.

Researchers

The third author, an AAC researcher and licensed SLP, acted as the interventionist in each session. She was not trained on study procedures because she developed the procedures for the study. In each session, either the first or fifth authors served as the “wizards” operating the context-aware prototype behind the scenes (see “Context-aware AAC Prototype” Materials subsection for more information). Training for the wizards involved a review of the intervention and data collection procedures as well as an overview of the technology used. All the researchers were white women with previous clinical experience providing intervention services to autistic children.

Materials

Traditional AAC Technology (Condition A): LAMP on a large iPad Pro (12.9” screen size) was the traditional AAC technology used in the current study. This app was selected because of the popularity of its use with autistic children, and its preset availability of a page depicting colors. It was also an app that all three participants had at least some exposure to outside the context of the study. At the beginning of each trial, the LAMP app was open to the preset colors page. When used in this study, the LAMP app was always navigated to the page of colors by the interventionist before the session began to eliminate any navigational demands on the participants. The colors page on the app showed a range of colors displayed by block coloring of a “blob” in the different cells of the grid.

Context-Aware AAC Prototype (Condition B): A prototype web app housed on the same large iPad Pro served as the technology for Condition B, the context-aware option. The app was developed in collaboration with AAC users, including those with developmental disabilities, and their communication partners utilizing human-centered design practices to promote responsiveness to user needs and dreams and accessibility (Dorrington et al., 2016). For more information about the prototyping and development approach, see Holyfield and colleagues (2024). The prototype emulated context-aware AI-powered technology that would, in response to a question and closed set of responses from a communication partner, offer the listed response options just-in-time to the individual using AAC. From there, the individual using AAC can select a response to the question from the response options provided and the response is spoken aloud.

The prototype operated on a “wizard-of-oz” design approach, which allows users to interact with experimental mock interface features that are actually human-controlled, but appear to be a fully developed feature of the technology (Maulsby et al., 1993). In this case, the “wizards” cued the color options listed by the interventionist while they were spoken aloud to make it appear as if AI visualized those choices on the AAC device automatically using speech recognition. For example, if the participant were coloring a fire truck on a sheet of paper, the interventionist may have asked, “Is the fire truck red, blue, purple, or green?”; the wizard would then cue red, blue, purple, and green squares to appear as a color block photo on the participant’s AAC device within one second of being selected. Figure 1 illustrates the view on both the “wizard’s” and the participant’s devices. This approach can potentially save time and resources required to develop fully functional AI-powered context-aware AAC technology by replicating the user experience early in the design process. It allows researchers to determine whether further investment is justified before committing to full-scale development. Additionally, this method enables primarily prelinguistic communicators to participate early on in a design process, which traditional design approaches often limit due to their reliance on extensive verbal responses (e.g., lengthy interview questions).

Unlike the LAMP app, the grid on the prototype the participants interacted with had only four large cells, each depicting a block of one color. Also, unlike the LAMP app, the prototype displayed only the four colors that were listed by the interventionist as they were listed by the interventionist. This approach was developed to promote accessible expressive communication from primarily prelinguistic communicators such as those who participated in this study by reducing visual-cognitive demands and aligning AAC with the immediate interaction environment (Holyfield et al., 2019; Holyfield, Light et al., 2025; Holyfield et al., 2024).

Art Supplies: Art supplies were used to support the context of the current study. Supplies included paint, paint brushes, markers, daubers, stamps, stickers, coloring books, sticker books, coloring pages, and blank paper.

Procedures

Participants engaged in 3-4 sessions over the course of two weeks until they completed all 20 trials across the two conditions. The number of trials varied per session due to the length

of the art activity but ranged between four and eight. All sessions occurred in the morning, one-on-one with the investigator and participant with the “wizard” present in the room to operate the context-aware prototype in condition B and collect data but not to engage in the interaction.

Conditions A and B: Session procedures were identical across phases and conditions except for the difference in technology used in the two conditions. Each session revolved around an interaction between the participant and the interventionist during art activities, following the lead of the participant. Each phase consisted of five trials that constituted the five data points for that phase. While engaging in the art activity, the interventionist asked five questions that served as the five trials. As the interventionist asked the question, the interventionist held the iPad in front of the participant at midline and eye level. Each question inquired about the color the child was currently using or just used within the art activity, and included a set of four-color choices with one of those options being the correct answer (e.g., asking, “What color is the starfish? Pink, purple, blue, or green?”) after the child had just colored the starfish purple. For each condition, the five trials each inquired about a different color. During Condition B (the context-aware AAC), the order of the correct color within the list spoken by the interventionist occurred pseudo-randomly so that each correct choice was in a different order than the correct choice from the previous trial.

In Condition A (the traditional AAC), the participants saw the color page of the AAC app as the interventionist asked them to label colors. In Condition B (the context-aware AAC), participants saw the corresponding four-color choices appear on the AAC app as they were being spoken aloud by the interventionist, activated through a different device onto the AAC screen by the “wizard” (see Figure 1). Only colors available on the traditional AAC app as choices were targeted as responses for either condition. After asking each question, the interventionist waited for a response, up to 30 seconds, with the iPad centered for the participant to respond. The interventionist praised responses but did not provide differential feedback or instruction of any other form. Instead of following a typical baseline and intervention format, this study focused on evaluating the context-aware AAC technology itself by comparing it to a traditional AAC. During this study, no instruction or correction was given to the participants in using the AAC in either condition.

Procedural Fidelity: Procedural fidelity was collected live by the session’s “wizard” using a checklist of the above steps for each trial. That is, the following steps were marked as completed or not completed in each of the five trials: (a) position AAC near participant’s midline and eye level, adjusting as needed to maintain this position throughout the trial; (b) pose a question related to color in the current activity with four color choices listed; (c) wait up to 30 s for a response; (d) praise responses, and (e) do not provide differential or corrective feedback. Procedural fidelity was calculated per block of 5 trials that constituted a condition block, creating a total of 25 total steps. To calculate procedural fidelity, the steps marked as completed each block were divided by 25 to yield a percentage. Procedural fidelity was collected across 100% of study sessions for all participants, and fidelity was a mean of 92.6% (range 83.3%-100%) across blocks of trials.

Data Collection

All data were collected live during study sessions by the “wizard”. The wizard used a stopwatch app on a phone that she started at the beginning of each response opportunity and stopped, recording the time, after a response or after 30 s of no response. The time was recorded in alignment with the five second time blocks outlined in the “Measure” section below – five seconds or less, ten seconds or less, fifteen seconds or less, twenty seconds or less, twenty-five seconds or less, or thirty seconds or less (with each exceeding time block excluding the time block before it). The wizard also recorded whether the response was accurate or inaccurate. These data served as the primary data in the study.

Measure: The dependent variable measured in the current study was accurate color label responses provided by the participants each trial. If the participant did not respond within 30 seconds of the question being asked in a trial, it was not counted as an accurate response. If the participant responded with a color other than the correct color to answer the question in a trial, it was also not counted as an accurate response. Accurate responses were coded in occurrence to the time in which a participant required to generate a response given the importance of quick responses in educational settings. Therefore, the following codes were possible for each trial: no correct response, accurate response requiring 30-26 seconds, accurate response requiring 25-21 seconds, accurate response requiring 20-16 seconds, accurate response requiring 15-11 seconds, accurate response requiring 10-6 seconds, or accurate response requiring 0-5 seconds. Responses would have been accepted through any modality (e.g., speech approximation, AAC use); however, participants in the current study only responded through use of AAC.

Interrater Reliability: All sessions were recorded on video to allow for the coding of the data for interrater reliability from a naïve person. All videoed sessions were reviewed by an undergraduate Communication Sciences and Disorders honors student naïve to the study, its goals, and hypotheses. The student was trained to code sessions for the study’s measure – labels from participants in response to trials. The student was trained using video of a similar interaction that was not a part of the study, then independently coded all study videos. Responses were coded as correct or incorrect, and responses were also coded for time to respond using the five second time blocks to group the data and allow for some flexibility in the starting/stopping of the stopwatch. Interrater reliability at the level of coding for an accurate response or no accurate response was 100%. Interrater reliability across all sessions and participants at the level of response time was 85.9% (range: 79.6% - 98.2%).

Data Analysis

Data were graphed in accordance with single-case experimental design standards and visually analyzed by the same standards (Kratochwill et al., 2013; Tate et al., 2016). Graphs were analyzed for each participant with respect to level, trend, slope, variability, and immediacy of effect across study phases. Effect size was measured using calculations of nonoverlap of all pairs (NAP; Parker et al., 2011).

Results

Figure 2 illustrates the data for each participant from each trial marked on the x-axis. The scale on the y-axis represents the amount of time left in the 30 s wait period in five second intervals. If a participant failed to respond or did not respond correctly, it is shown at the bottom of the axis in the zero position. Responses that occurred at the end of 30 second wait period (or with only 5-0 seconds remaining) are shown at the next level. The scale progresses in speed of response up the y-axis so that responses that occurred the fastest or as soon as the question was asked (with 30-26 seconds remaining of the wait period) are at the top of the graph. This scale was used so that it could be easily seen which condition produced faster responses as this would be important in classroom settings.

Overall, all three participants showed higher levels of accurate responding using the context-aware AAC versus the traditional AAC. Collectively, all three participants responded accurately in only 20% of sessions using the traditional AAC and 80% of sessions using the context-aware AAC. Furthermore, all participants generally responded more quickly (or with more time remaining during the allotted wait period) using the context-aware AAC than the traditional AAC. Two of the participants received NAP scores that reflected a large effect size while the remaining participant's score reflected a moderate to large effect size. Collectively, participants' data had a NAP score of 0.827, which indicates a moderate to large effect size (Parker et al., 2011).

Anna

In the first traditional AAC condition, Anna did not respond during any of the five trials. In contrast, during the first context-aware AAC condition, Anna responded accurately to each question within 5 s of the instructor asking the question. Therefore, she had 30-26 seconds left in the allotted wait period when the question was asked. During the second traditional AAC condition, Anna failed to respond during the first two sessions. However, she responded for the next three sessions with 15-11 seconds, 5-0 seconds, and 16-20 seconds, respectively, left in the wait period. During the second context-aware AAC condition, Anna responded with 30-26 seconds left in the wait period for four out of five trials and with 10-6 s left for the remaining trial. Overall, Anna responded accurately during only 30% of trials while using the traditional AAC but responded accurately during 100% of trials using the context-aware AAC.

Anna's data represent a low level with moderate variability during both traditional AAC conditions and a comparatively high, stable level with minimal variability during both context-aware AAC conditions. Although her data show an increasing trend during the second traditional AAC condition, they do not reach the same level as either context-aware AAC condition. In addition, there is a high degree of immediacy between the first traditional AAC condition and the first context-aware AAC condition. Her data immediately drop back down during the second traditional AAC condition and immediately increase again with the second context-aware AAC condition. Additionally, Anna's data has a NAP score of 0.975, which represents a large effect size (Parker et al., 2011).

Blake

Similar to Anna, Blake did not respond during any of sessions under the first traditional AAC condition. However, he responded with 21-30 seconds left during four out of five trials in the first context-aware AAC condition. During the fifth trial, Blake responded with the incorrect color. Under the second traditional AAC condition, he did not respond for three out of five trials. During the remaining sessions, he once responded with the incorrect color and once responded accurately with 10-6 s remaining of the allotted wait period. In contrast, during the second context-aware AAC condition, he responded accurately during four out of five trials with 15-11 seconds remaining for two trials, 0-5 seconds for one trial, and 25-21 seconds for one trial. During the second trial, Blake responded with the incorrect color. Overall, Blake responded accurately during only 10% of trials using the traditional AAC and 80% of trials using the context-aware AAC. In terms of responding, Blake responded during 100% of trials using the context-aware AAC, however two of these responses were inaccurate. Whereas Blake only responded during 20% of trials using traditional AAC and one of these responses was inaccurate.

During both traditional AAC conditions, Blake's data demonstrate low, stable levels of responding. In contrast, his data during both context-aware AAC conditions demonstrate an overall higher level of responding with more variability and an increasing trend during the second context-aware AAC condition. In addition, his data demonstrate immediacy during both changes between traditional AAC and context-aware AAC. Blake's data received a NAP score of 0.88 which represents a moderate to large effect size (Parker et al., 2011).

Calvin

Calvin responded with 20-16 seconds remaining of the allotted wait period during only one of the first five sessions under the traditional AAC condition. He responded inaccurately during three trials and did not respond at all during the remaining trial. Under the first context-aware AAC condition, he did not respond during the first trial and responded inaccurately during the third trial. He responded accurately with 25-21 seconds, 20-16 seconds, and 0-5 seconds remaining for the other three trials. During the second traditional AAC condition, he responded inaccurately during one trial and did not respond during three trials. However, he responded accurately with 26-30 seconds remaining during one trial. For the second context-aware AAC condition, he responded inaccurately during one trial and did not respond during another trial. For the remaining three trials, he responded accurately with 10-6 seconds left on two trials and 20-16 seconds left on one trial. Overall, Calvin responded inaccurately during four sessions and did not respond for four sessions using traditional AAC, which means he answered accurately for only 20% of sessions. In contrast, Calvin responded inaccurately during two sessions and did not respond during two sessions using the context-aware AAC, meaning he responded accurately during 60% of sessions.

During both traditional AAC conditions, Calvin's data show relatively low, stable levels of responding. His data during both context-aware AAC conditions demonstrate more variability, but with an overall higher level. His data show a steady upward trend during the second context-aware AAC condition. His data do not demonstrate any immediate change

between conditions. However, the NAP score for Calvin's data was 0.66, which indicates a moderate effect size (Parker et al., 2011).

Discussion

The purpose of this study was to compare the effects of using a context-aware AAC prototype with automated response options to a traditional AAC with topic grid display on the academic communication responses of three young autistic children. The results demonstrated that all three participants were able to communicate the answers to academic questions accurately at least 80% of the time using the context-aware AAC compared to 20% of the time using the traditional AAC. Furthermore, all participants generally responded more quickly using the context-aware AAC. NAP scores indicate a moderate to large effect size for all participants (Parker et al., 2011).

These results are consistent with previous research, demonstrating that context-aware AAC may increase the ease and accuracy of communication for young children with autism spectrum disorder (Holyfield et al., 2024; 2025). As such, this study provides additional evidence toward the promise of context-aware AAC to support participation and communication (Kane et al., 2012). Furthermore, it adds to the literature base demonstrating that AAC interventions may be useful for a variety of communication functions, including academic responding (Griffen et al., 2025; Yorke et al., 2018). Because this study replicated a natural interaction that takes place between teachers and students in the classroom every day, it provides one example of how AAC intervention might be more easily implemented in a classroom setting if teachers had access to more efficient technology, such as a fully functional context-aware AAC that allowed for automated JIT programming. Technology with such functionality may better meet the needs of children who require AAC (Lorah et al., 2024; Williams & Holyfield, 2025) and may therefore better support language learning and social participation at school (Holyfield, Light et al., 2025; Light et al., 2025).

Clinical Implications

Enhanced Responsiveness and Classroom Assessment: One significant finding of this study is that not only did participants have more accurate and timely responses, but they also responded more frequently using the context-aware AAC, even when those responses were incorrect. In a classroom setting, incorrect responses may provide teachers with more information about the type of skills the student is missing and what lessons are still needed (Käfer et al., 2019). For example, if a student responds incorrectly every time red and orange are presented, but responds correctly with blue, green, and yellow, this might indicate that the student has trouble discriminating between only two colors. Whereas, if a student fails to respond, the teacher is left without knowing if the student is unable to discriminate between any of the colors or lacks the communication skills to demonstrate their knowledge. This distinction was highlighted in this study when considering Anna's and Blake's responses. During the first traditional AAC condition, neither Anna nor Blake responded to any of the interventionist's questions, which might leave one to assume that they could not discriminate any of the colors. However, when the context-aware AAC was introduced, they both responded with 80% or greater accuracy, which indicates that they did

have the ability to discriminate between colors, but lacked the ability to communicate those responses using traditional AAC technology.

Impact on Instructional Planning: In a classroom setting where teachers must constantly assess their students' academic and cognitive skills in many areas, it is vital that students are able to communicate the skills they possess so that teachers will have more opportunities to provide feedback, scaffold responding, and tailor lesson plans appropriately. When AAC users are in the classroom, teachers should consider how the limitations of technology may impact their students' ability to demonstrate their skills. Equipping students with more effective and efficient communication tools may increase overall responding, allowing children with complex communication needs to have more educational opportunities and be more engaged in classroom activities.

Expanding Communication Functions Beyond Requests: This study provides evidence that children can learn communication for a variety of functions using AAC. Both Blake and Calvin had inconsistent access to AAC prior to the start of this study and yet both were able to communicate the answers to questions throughout the study. This demonstrates that AAC intervention for children learning language does not need to be limited to requesting needs and wants but should be expanded to include other functions of communication (Holyfield et al., 2025). When designing instructional programs for young autistic children, practitioners should consider incorporating communication targets across a wide range of functions that will support more engagement and participation in academic and educational settings.

Innovation and Collaboration in AAC Development: This study also illustrates the need for continued innovation in AAC, particularly in the area of developing technologies for automated augmentative input that may allow AAC users to respond more quickly and accurately. Increased collaboration in the field between researchers, teachers, AAC users, and technology experts is needed to develop practical tools that can help autistic children in a variety of settings, including academic classrooms. Teachers would benefit from technologies that reduce the amount of time and effort required by communication partners for device programming so that they could focus on instructional time with their students (Light et al., 2025).

Ethical Considerations in Emerging AAC Technologies: As technologies become more advanced and available, professionals will need to consider a variety of ethical issues. Technologies such as context-aware AAC that automate responses do not eliminate the need for professionals to ensure that their clients receive safe, individualized, and developmentally appropriate care. Professionals are responsible for ensuring that client data are protected and stored appropriately. They will need to monitor systems for racial and gender biases that are common in AI (Cai et al., 2022). Perhaps most importantly, they will need to focus on ensuring personalization of context-aware AAC and monitor programs for use of developmentally appropriate vocabulary and communication targets. Another ethical consideration for both professionals and technology developers will be to maintain authenticity of communication for AAC users. Automated technologies, such as

context-aware AAC, do not replace the need for professionals, such as speech-language pathologists, teachers, and behavior analysts, but rather, they can support professionals and caregivers to provide more effective communication tools to their clients (Light et al., 2025).

Promoting Self-Determination Through AAC: These innovations are not only potential pathways for enhanced efficiency in communication, but they are also critical avenues to explore for building self-determination among AAC users. By enabling people who use AAC, including young autistic children, to express preferences, make choices, and self-advocate in real time, context-aware systems can promote autonomy and agency across environments. In this way, technology development in AAC should be guided not only by functionality and ethics, but also by a commitment to enhancing opportunities for users to direct their own lives and participate in their communities in the ways they choose to. Prior research highlights that self-determination is central to improved academic, social, and postschool outcomes for students with disabilities (Raley et al., 2021; Shogren et al., 2015), underscoring the importance of embedding these opportunities within emerging technologies. As AAC continues to evolve, framing innovation through the lens of self-determination will ensure that technology serves as both a communication tool and a vehicle for empowerment by centering agency for AAC users as the driver for design-based decision making (Raley et al., 2024).

Limitations

This study was not without some limitations. First, all three participants started in the traditional AAC condition, which may have impacted the context-aware condition that followed. All participants had previous experience with traditional AAC so it might serve as a better baseline, or standard, of performance. Both conditions were repeated to decrease the likelihood of sequence effects. However, it is still possible that the sequence of the conditions had an impact on the observed effects (Ledford & Gast, 2024).

A second limitation is that the participants represent a very limited age range as they were all between four and five years old. Although this represents a pivotal age for development in language and communication skills, when examining the application of context-aware AAC to academic skills, a broader age range of participants would be beneficial. Future studies should examine its use for older children in junior high and high school. Also, this study focused only on academic communication and did not include back-and-forth conversations or peer interactions that are essential for AAC users. Future studies should explore the use of context-aware AAC for social conversations using peers, siblings, or family members as communication partners.

This study was also limited in that it did not implement an intervention to teach the participants academic skills. In other words, although the participants may have learned to identify some of the colors through the reinforcing effects of the interventionist's praise, that was not the main purpose of the study. This study only examined the participants' access to expressing information they already knew. Although Calvin demonstrated an increase in accurate responding using the context-aware AAC, he still had more incorrect responses in both conditions than the other two participants. This may indicate that he had fewer

discrimination skills than the other two participants and may need further intervention. Future studies should look at using context-aware AAC to teach new knowledge of foundational academic skills to young autistic children. The intervention implementation was also limited by some sessions having lower intervention fidelity than others. This creates a limitation for the internal validity of the study, though the technology accessed in each condition – the independent variable in the study – was clearly demarcated.

Although many adult AAC users have advocated for the development of context-aware AAC, this study did not include measures of social validity or a choice condition to evaluate participants' preferences (Lorah et al., 2024). This omission is a notable limitation because AAC interventions should adapt to the needs and priorities of individuals who use AAC (Holyfield, Light et al., 2025). Future studies could address this oversight by adding a final phase that examines participant choice or by incorporating measures such as indices of happiness to gauge participants' acceptance and enjoyment of the procedures and technology (Carnett et al., 2022).

Lastly, the study was limited by the technology itself. As previously stated, the authors were using a prototype and not a fully functioning context-aware AAC. Moreover, there were many features of the AAC device that differed between the two conditions, such as the number, size, and style of the icons. While these differences may have contributed to the improved performance in the context-aware AAC condition, they also represent an inherent advantage of context-aware AAC, which is able to identify relevant information from the environment and the communication partner and filter out unnecessary options, reducing the number of icons on the screen and allowing icons to be bigger if needed. By reducing the cognitive load for the AAC user, they may be able to respond more quickly and accurately. It is impossible, and perhaps unnecessary, to determine how each feature of the context-aware AAC contributed to making it more effective during this study.

Conclusion

Despite these limitations, this study demonstrates that context-aware AAC may lead to more frequent, accurate, and faster responding for young children on the autism spectrum when compared to traditional AAC. Furthermore, this study provides evidence that thoughtfully designed AAC technology can support young children on the autism spectrum in demonstrating their academic knowledge more effectively. Teachers may find technological advances, such as automated augmented devices, helpful in reducing the time and effort demands of traditional AAC programming and in providing more accurate assessments of students' skills. Advances in AAC technology may help young students to communicate more effectively and efficiently, and thus, be able to engage and participate more fully in academic settings. Researchers should consider developing these technologies collaboratively with AAC users, technology experts, teachers, and other valued constituents (Holyfield et al., 2025; Williams & Light, 2025).

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Data Availability

All data generated or analyzed during this study are included in this published article.

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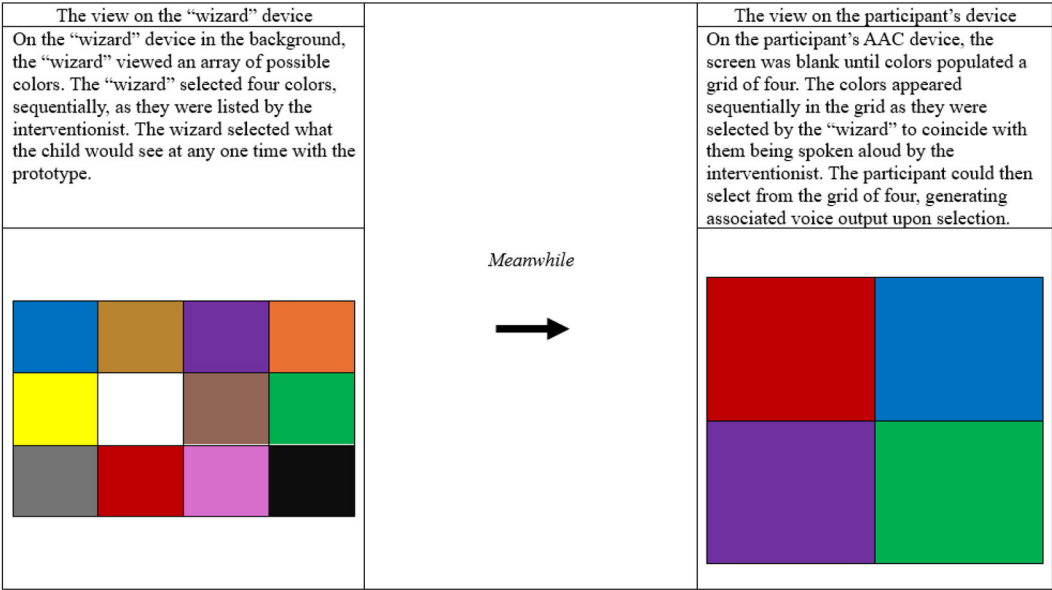
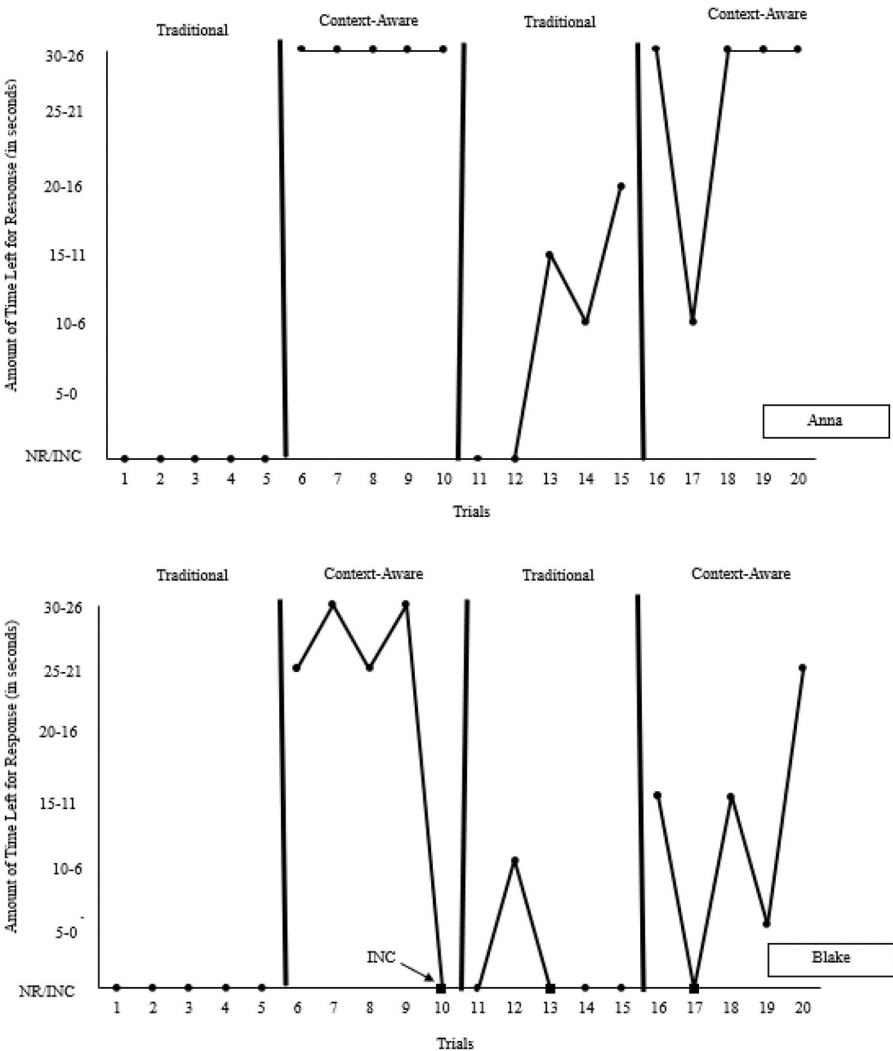


Figure 1. “Wizard” and Participant Device Displays
Note. Illustration mimicking the device screens as they were seen by the “wizard” and the participant with accompanying description of use.



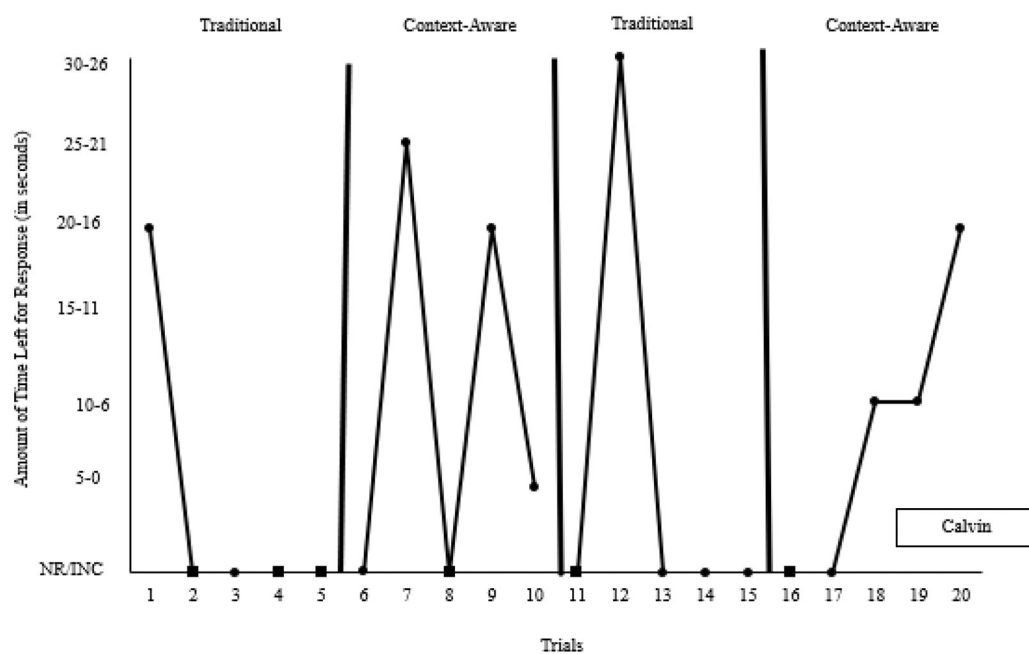


Figure 2. Participant Responses

Note. Squares represent incorrect responses.

Table 1
Participants' Demographic Information

Participant	Age (months; years)	Sex	Race	Primary Diagnosis	Primary Communication Modes ^a	Unique Words Expressed Regularly ^a	Basic Color Knowledge ^a
Anna	5;0	Female	White	Autism Spectrum Disorder	Prelinguistic behaviors; AAC app on a tablet; small set of vocal approximations	<25	100% (8/8)
Blake	3;10	Male	White	Autism Spectrum Disorder	Prelinguistic behaviors	<10	75% (6/8)
Calvin	5;8	Male	White	Autism Spectrum Disorder	Prelinguistic behaviors	<10	100% (8/8)

^aBased on professional report and screening